Hsuan-Hui Vankley Yi

Evanston, IL vankleyyi@u.northwestern.edu Portfolio LinkedIn GitHub

EDUCATION

M.S. in Robotics *Sep 2026 – Dec 2027 (Expected)*
Northwestern University, Evanston, IL

B.S. in Industrial Engineering and Engineering Management *Sep 2021 – Dec 2025*
National Tsing Hua University, Taiwan

SKILLS

Robotics	ROS 2, RealSense, CoppeliaSim, Modern Robotics, DJI Drone SDK
Programming	C#, Python, C++, C, Linux, Git, Java
VR & XR	Meta Quest, Unity, Blender, HoloLens, Humanoid Rigging
Hardware	Microcontroller Programming, CAD (Creo, SolidWorks), KiCad, Arduino, ESP32, Teensy, FPGA, OpenBCI Galea, Haptic Feedback
Languages	English (Professional), Mandarin Chinese (Native), French

EXPERIENCE

Haptics Research Intern, ISIR, Sorbonne Université & CNRS, France *Jan 2026 – Jun 2026*
- Investigated mechanical structures and silicone material deformation for stretch-based stiffness feedback, designing and prototyping an electrically actuated mechanism.
- Developed a Hall-effect sensing system to track soft-cube manipulation and capture object deformation and interaction.

Mechatronics Intern, Logitech, Taiwan *Mar 2025 – Aug 2025*
- Built analog circuits, integrated Hall-effect sensors, and developed firmware with real-time PC–hardware communication, completing full system integration from PCB to mechanical design.
- Worked on FPGA-based ultrasonic haptic prototypes, experimenting with custom waveforms for LRA and piezo actuators.
- Collaborated with international teams and presented at the headquarters, earning recognition for innovation.

VR Game Development Contractor, FishForest, Remote *Jul 2024 – Nov 2024*
- Built a VR experience based on field interviews and observations to translate fishing-community culture into an interactive underwater environment. Applied immersive technology to support local revitalization in a rural setting.

Game Development Intern, Blacksmith Technology Limited, Taiwan *Sep 2024 – Nov 2024*
- Focused on the game design of a sports training game while fine-tuning an LLM chatbot.

PROJECTS

Pen Picker – Robotic Pen Retrieval System *Sep 2026*
- Implemented vision-based pen detection and 3D localization with Intel RealSense RGB-D data and OpenCV, guiding a robotic arm for grasping.

Haptic Device Development In Logitech *May 2025 – Aug 2025*
- Explored haptic feedback technologies through hands-on work with electromechanical actuators, sensor integration, embedded firmware, and hardware-software interfacing.

EmBelt – Wearable Haptic XR System *Sep 2023 – Jan 2025*
- Designed a wearable haptic feedback system integrating force-sensitive resistors and DC motors with Unity-based VR to explore body image awareness through interactive sensory experiences.
- Received 2 awards and exhibited at Laval Virtual 2024, France, and SIGGRAPH 2024 Emerging Technologies, USA.

Helmo – Wearable Haptic System *Aug 2025*
- Developed a wearable haptic system using ERM motors to alert scooter riders to vehicles in their blind spots through directional vibrations, enhancing safety without visual or audio distractions.

Hand Gesture Controlled Drone Multi-View Interface Evaluation *Jul 2024 – Dec 2024*
- Created a Unity-based drone interface and conducted a user study comparing first- and third-person perspectives to examine their effects on user interaction.

Sizzler! – Brain-Computer Interface for XR Interaction *Jan 2024*
- Connected the OpenBCI Galea headset to an immersive XR environment to explore brain-signal-driven interaction.
- Received Meta Honorable Mention at MIT Reality Hack 2024, USA.

Home, Fish Farm – Educational VR Experience *Jul 2024 – Nov 2024*
- Transformed local aquaculture practices and ecosystems into an interactive virtual tour in Unity.